

FraudShieldAI - Links, Commands & Profiles

1. Running the API Server

To start the FastAPI server, use the virtual environment where all dependencies are installed (.venv_new). Run the following command from the project root:

Command:

```
.\.venv_new\Scripts\python -m uvicorn 09_api:app --reload
```

This will start the server at <http://127.0.0.1:8000>

2. Web Interfaces (Open in Browser)

Once the server is running, you can access the following web interfaces:

Spoof UPI Payment App:

<http://127.0.0.1:8000/>

Bank Analyst Dashboard (with GNN Network Visualization):

<http://127.0.0.1:8000/console>

API Documentation (Swagger UI):

<http://127.0.0.1:8000/docs>

3. Simulation Note & Profile Specifications

NOTE: This application is an interactive simulation UI. Payment transfers map user profiles to actual precomputed fraud scores sampled from real transactions in the IEEE-CIS Fraud Detection dataset (590,540 real transactions).

User Profiles:

- Faris (faris): Regular Personal Account
- Rahul (rahul): Frequent Peer Transfers
- Ahmed (ahmed): Retail Merchant Account
- Priya (priya): Corporate High-Volume
- Ananya (ananya): Freelance / International
- Arjun (arjun): New Account (Low Behavioral History)
- Kiran (kiran): Whitelisted E-Commerce Entity
- Neha (neha): High-Velocity Account

4. API Endpoints (cURL / Test Commands)

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You can test the API endpoints directly using PowerShell:

Health Check:

```
Invoke-WebRequest -Uri "http://127.0.0.1:8000/health" | Select-Object Content
```

Get Fraud Score by Transaction ID:

```
Invoke-WebRequest -Uri "http://127.0.0.1:8000/predict_by_id?transaction_id=3012474" | Select-Object Content
```

Simulate a Payment (Sender, Receiver, Amount):

```
Invoke-RestMethod -Method Post -Uri "http://127.0.0.1:8000/pay" -Headers @{"Content-Type"="application/json"} -Body '{"sender":"faris","receiver":"rahul","amount":100}'
```

5. Running the Latency Benchmark

To verify the sub-2-second response time requirement, run the latency benchmark script. Ensure the server is running before executing this.

Command:

```
.\.venv_new\Scripts\python benchmark_latency.py
```